

SAGAR SINGH BHOLA

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EDUCATION

University of Michigan, Ann Arbor

Expected Dec 2025

Bachelor of Science in Computer Science

Major GPA: 3.8

- **Relevant Coursework:** Data Structures and Algorithms, Computer Networks, Computer Security, Web Systems, User Interface Development, Human-Centered Software, Probability & Statistics

TECHNICAL SKILLS

Languages: Python, C, C++, Java, HTML, CSS, JavaScript, SQL

Frameworks/Tools: React, Next.js, Flask, Node.js, Docker, AWS, Vue, Git, Postman, Figma, Pandas, Selenium

EXPERIENCE

Software Engineer Intern

May 2025 – Aug 2025

Ciena Healthcare

Southfield, MI

- Automated generation of 50+ Michigan Medicaid cost reports across skilled nursing facilities using Python, Pandas, and Selenium, decreasing manual workload by **90%**.
- Engineered an internal dashboard with Next.js + Flask to display monthly and year-to-date financial summaries, accelerating reporting turnaround and improving data accessibility for finance teams.
- Implemented CI/CD pipelines to automate testing and deployment of scalable Python data processing scripts, increasing reliability and efficiency.

Software Engineer, Co-Lead

Mar 2025 – Present

Yes, No, Fingers (U-M Language Resource Center)

Ann Arbor, MI

- Architected a secure, multilingual medical terminology translation app powered by a fine-tuned M2M100 and RAG pipeline, enabling high-accuracy translations across 40+ languages for Detroit Free Clinic patients and clinicians.
- Developed multilingual text-to-speech, implemented advanced search, and a secure backend API with user-friendly frontend, ensuring fast and reliable access to medical terminology in clinical settings.
- Maintained HIPAA compliance by filtering sensitive information and localizing all processing, ensuring privacy by preventing external data exposure throughout translation workflows.

PROJECTS

Payflow[OCR] – AI Invoice Processor | Next.js, Python, Flask, SQL, AWS

Aug 2025 – Present

- Designed an end-to-end AI-driven automation platform for enterprise accounts payable, cutting manual invoice processing time by **80%** and directly accelerating month-end financial close.
- Automated invoice intake via mail forwarding and implemented dynamic multi-level approval workflows, enabling secure and auditable processing for cross-functional finance teams.
- Built robust, scalable data extraction pipelines to auto-populate General Ledger distributions and generate integration-ready exports, streamlining onboarding for diverse financial systems.

Healthcare Translation App | React, Flask, Python, Google Gemini

Jan 2025

- Developed real-time web app translating medical terms into 10+ languages with audio playback for non-English-speaking patients.
- Utilized LLMs and AI for translation verification, ensuring medical accuracy and reducing miscommunication.
- Collaborated with a cross-functional team of 5 in weekly agile sprints to deliver and deploy new features.

Instagram Clone | React, Flask, Python, SQL, AWS

Oct 2024

- Built a performant social platform with infinite scroll, secure user authentication, and dynamic post feeds using React and Flask.
- Optimized REST API architecture and reduced page reload time by **90%** through asynchronous JavaScript and server-side rendering.

LEADERSHIP EXPERIENCE

Technology Lead – Cyber Threat Competition, National Finalist

Feb 2023

Deloitte Risk & Financial Advisory

Dallas, TX

- National finalist, ranked top 10% of 1,500+ individuals by leading incident response strategy and presenting solutions to Deloitte cybersecurity executives.